

H2GUARD · DESIGN PRECIS

H2Guard design precis

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DOCUMENT CONTROL

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REVISION HISTORY

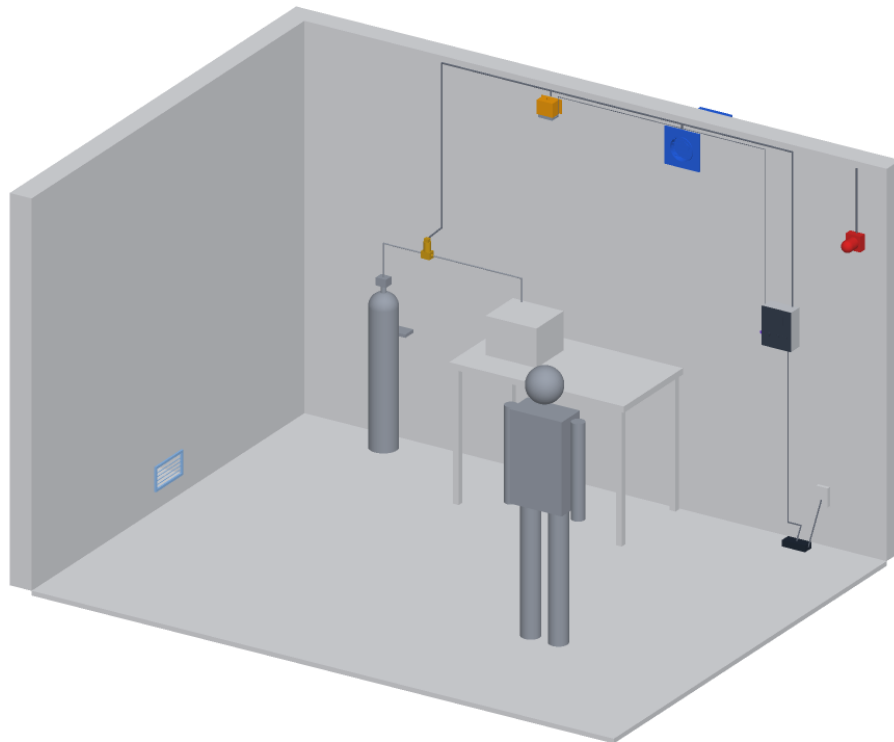
Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); timed escalation firmware rule, head placement rule, budget \$265, R13 scope; drawing HGD-DWG-001 to Rev P2
0.3	2026-09-25	Amish Chadha	TRL 3 update; design choices adopted for TRL 3 per HGD-DDR-001 (pending Amish's review); numbers from HGD-CAL-001; fan re-specified, thinner arrestors, bump test cup and port added; series trip relay; parametric model and drawing HGD-DWG-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, components, first-order numbers, design choices, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

H2Guard design precis

H2Guard protects one small room where hydrogen is used. A detector head at the ceiling carries a catalytic sensor that reads 0 to 100 % of the lower flammability limit (LFL) and a metal oxide sensor for early warning. A wall controller at chest height keeps a 150 mm exhaust fan running at high level, holds a normally closed solenoid valve on the hydrogen supply open only while all is well, and drives a sounder and beacon. At 10 % LFL it warns and boosts the fan; at 25 % LFL it closes the valve and latches the alarm. Loss of power, a sensor fault, a stopped fan or a crashed controller all close the valve, and a hardware comparator opens a relay in series with the valve without the firmware. The TRL 3 calculations (HGD-CAL-001) give 35.7 s from a leak to the valve closing with an assumed 30 s sensor t_{90} , about 5 % LFL in the 30 m³ room for the 5 L/min design leak with the fan running, and a parts cost of \$264 against the \$265 budget. A warning held for 5 min also closes the supply, so a leak that the head sees off the plume axis is stopped within about 6 min. H2Guard is a research and teaching prototype, not a certified gas detection system.

H2Guard

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale. Grey room, bench, gas store, supply line, cables and bump test tube are context.

Figure 1. H2Guard in the 30 m³ reference room, with a 1.75 m person for scale. Detector head (orange) above the bench apparatus, exhaust fan (blue) high on the back wall, make-up air grille low on the far side wall, solenoid valve (gold) on the supply line, controller with its bump test port, and beacon near the door. Grey parts are context. Built from the parametric model.

■ How it works

1. **Sense.** Hydrogen rises from a leak as a buoyant plume and spreads under the ceiling. The detector head sits directly above the likely leak point, with its sensor ports 175 mm below the ceiling. The placement rule (HGD-DDR-002) puts a head over each likely leak point, with a second head where leak points are far apart; the plume is only about 141 mm in radius at the ports, and the allowed horizontal offset is to be set from a measured plume width at TRL 4. The catalytic sensor measures 0 to 100 % LFL and sets the warning and trip. The metal oxide sensor responds from about 30 ppm and shows small, slow leaks as a trend long before the catalytic reading moves; it never trips the system alone, because humidity and other vapors affect it. Each sensor sits about 2 mm behind a 2 mm sintered arrestor disc, which adds about 1.5 s of lag.
2. **Decide.** The controller reads both sensors. The microcontroller handles set points, display, logging and self-test, and switches the valve through a MOSFET. In parallel, a hardware window comparator watches the catalytic sensor bridge: above the trip level, or if the bridge reads open or shorted, it sets a hardware latch that opens a relay in series with that MOSFET. Either switch opening closes the valve, and only the key resets the latch.
3. **Act.** - **Normal:** fan at continuous speed, valve energized (open), green light. - **Warning, 10 % LFL (0.4 % vol):** fan to boost, amber light, intermittent sounder. Clears itself when the reading falls. If the warning is held for 5 min, the microcontroller closes the valve and latches as for a trip (timed escalation, HGD-DDR-002). This is a firmware rule; the hardware comparator still acts only at 25 % LFL. - **Trip, 25 % LFL (1.0 % vol):** valve de-energized (closed), fan on boost, continuous sounder and red beacon. Latched: the supply stays closed until someone turns the reset key and the reading is below 10 % LFL. - **Fault:** any loss of 24 V power, sensor fault, heater failure, watchdog timeout (1 s) or fan stop (no tachometer pulses for 10 s) closes the valve and shows the fault.
4. **Ventilate.** The fan runs all the time, so it is never switched on in a flammable mixture, and so no gas can flow unless the room is being ventilated. Make-up air enters through a low grille on the far side of the room, sweeping the room toward the high exhaust.
5. **Record and test.** The controller logs readings, warnings, trips, faults and resets. A test button runs the sounder, beacon, fan boost and valve close. For a bump test, certified 1 % vol hydrogen span gas is connected to a capped port beside the controller; a 4 mm tube carries it up the cable route to a nozzle in the drip skirt under the head, which acts as a test cup. The reading settles in about 67 s. Because 1 % vol is the trip level, a full-span bump test also trips the system and proves the chain.

H2Guard: detect, decide, act chain for the design leak (HGD-CAL-001 estimates, 30 m3 room)

CONCEPT, NOT FOR FABRICATION



Figure 2. Detect, decide and act chain for the design leak in the 30 m3 reference room, with values from HGD-CAL-001. They are estimates; the sensor t90 is an assumption not yet confirmed from a datasheet.

■ Main components

Table 1. Main components. Numbers match bom/bom.csv , Figure 3 and drawing HGD-DWG-001.

#	Component	Choice	Notes
1	Detector head enclosure	ABS or polycarbonate box 110 x 80 x 90 mm, two 26 mm sensor ports facing down	Directly above the source, ports 175 mm below the ceiling
2	Catalytic hydrogen sensor	Catalytic (pellistor) sensor with hydrogen response, 0 to 100 % LFL, for example Figaro TGS6812	Primary trip sensor; needs oxygen and can be poisoned by silicones
3	Metal oxide hydrogen sensor	About 30 to 3,000 ppm, for example Figaro TGS2616-C00	Early warning and trend only
4	Flame arrestor discs and bump test cup	Sintered stainless discs 25 x 2 mm, pores 50 µm or less; drip skirt 96 x 66 x 22 mm forming the test cup	Not a certified flameproof assembly
5	Controller enclosure	IP65 polycarbonate wall box 200 x 250 x 90 mm	Top 1,025 mm below the ceiling
6	Controller board	RP2040-class microcontroller with 4 MB flash; comparator with hardware latch and series relay; drivers for valve, fan and alarm; tachometer input; event log	Firmware beyond a labeled sketch is TRL 4 work
7	Front panel	OLED display, key-switch reset, test button, status lights	The key keeps students from clearing a trip
8	24 V DC power supply	Certified plug-in supply, 24 V, 60 W	Only mains part; bought certified
9	Exhaust fan	150 mm mixed-flow EC duct fan, 24 V, about 450 m ³ /h free air, 200 Pa shut-off, tachometer; grille, sleeve, backdraft shutter, weather hood	Re-specified at TRL 3; runs continuously at about 43 % speed
10	Make-up air grille	Low-level wall grille 300 x 160 mm with insect mesh	Far side of the room from the fan
11	Normally closed solenoid valve	1/4 in, direct acting, brass, FKM seals, 24 V DC, 0 to 10 bar, Zener clamp	After the regulator; closes on loss of power
12	Sounder and beacon	24 V, about 90 dB at 1 m, red flashing	By the door, top 450 mm below the ceiling
15	Bump test port and tube	Capped push-fit port beside the controller, about 3 m of 4 mm tube to the cup nozzle	New at TRL 3 for R12

H2Guard: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Detector head enclosure
- 2 Catalytic H2 sensor, 0 to 100 % LFL
- 3 MOS H2 sensor, early warning
- 4 Flame arrestors and bump test cup
- 5 Controller enclosure
- 6 Controller board, series trip relay
- 7 Front panel: display, key reset, test
- 8 24 V DC supply, certified
- 9 Exhaust fan, 150 mm mixed flow
- 10 Make-up air grille, low level
- 11 NC solenoid valve, 24 V DC
- 12 Sounder and beacon
- 15 Bump test port (tube shown grey)

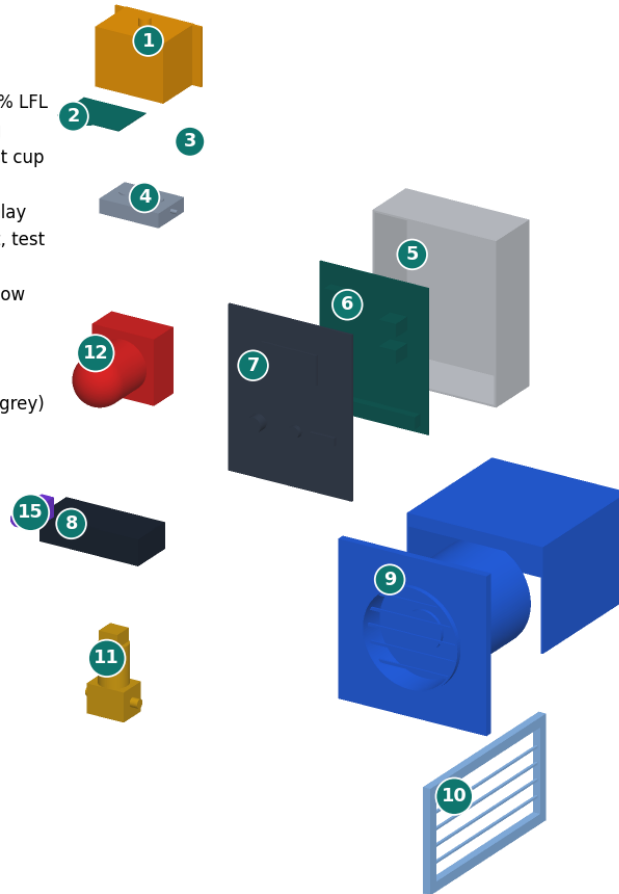
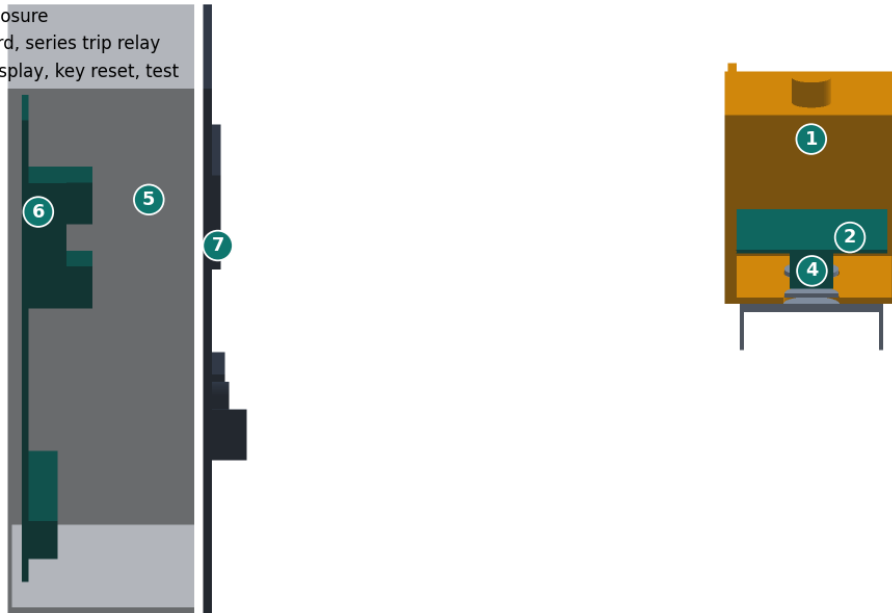


Figure 3. Exploded view of the H2Guard parts with BOM numbers. Cabling (BOM line 13) and fixings (line 14) are not shown.

H2Guard: section through detector head (right) and controller (left)

CONCEPT, NOT FOR FABRICATION

- 1 Detector head enclosure
- 2 Catalytic H₂ sensor, 0 to 100 % LFL
- 4 Flame arrestors and bump test cup
- 5 Controller enclosure
- 6 Controller board, series trip relay
- 7 Front panel: display, key reset, test



Head shown beside the controller; installed with its ports 175 mm below the ceiling. MOS sensor (3) is behind the cut plane.

Figure 4. Sections through the controller (left) and the detector head (right), each cut on a vertical plane and seen from the side. The head section passes through the catalytic sensor, its arrestor disc and the bump test cup. The head is drawn beside the controller for this view.

The general arrangement drawing [HGD-DWG-001](#) (Rev P2) gives the mounting heights and main dimensions from the parametric model `cad/src/model.py`. The blueprint concept sheet ([PDF](#)) and the [interactive 3D model](#) show the parts in place.

■ Key numbers

All values are from HGD-CAL-001, which gives the method and assumptions. They are first-principles estimates.

Table 2. Key numbers for the 30 m³ reference room and the 5 L/min design leak.

Quantity	Value
Hydrogen from a 100 W electrolyzer	0.39 L/min
Ceiling layer (0.3 m) to 25 % LFL without ventilation	7.2 min (whole room mixed: 60 min)
Room and upper layer with 150 m ³ /h exhaust	5.0 % LFL (2.5 % LFL on boost)
Plume at the detector head	About 15 % LFL mean, about 30 % LFL on the plume axis
Leak to valve closed	35.7 s with an assumed 30 s sensor t ₉₀ ; 0.17 s from trip to closed
Hydrogen released before closing	3.0 L, plus 0.47 L in the line downstream
Fan delivery against duct losses	347 m ³ /h at full speed (the TRL 2 fan gave 237 m ³ /h)
Inventory limit	300 L at 1 atm (25.1 g); a 0.13 mm orifice at 10 bar gauge limits a failure to 5 L/min
Power	13.1 W normal, 48.3 W warning, 40.3 W trip
Bump test	About 67 s; 1.36 L of span gas per test
Timed escalation, off-axis design leak	Valve closed at 5.6 min, 28 L released, room about 1.9 % LFL
Parts cost	\$264.00 against the \$265 budget (without fan and grille \$206)

Two findings matter for the design. First, the design leak trips the system only if the head is on the plume axis; beside it, the head sees about 15 % LFL and warns without closing the supply, although the room stays at about 5 % LFL. Second, a fan rated at the boost flow in free air cannot deliver it through a real duct, so the fan is re-specified. Amish accepted two remedies for the first on 2026-09-25 (HGD-DDR-002): the timed escalation and the head placement rule, both now in the design.

■ Key design choices

The choices below are decided by Amish, 2026-09-25: go with recommendation (HGD-DDR-001, confirmed by HGD-DDR-002).

- 1. Catalytic sensor for the trip, metal oxide sensor for early warning (D1).** Options were (a) catalytic plus metal oxide (about \$50 of sensors); (b) a certified molecular property spectrometer sensor (about \$249, t₉₀ under 20 s, poisoning resistant, [NevadaNano](#)); (c) metal oxide only. Adopted: (a) for the teaching prototype, with (b) documented as the upgrade for any room used without close supervision.
- 2. Normally closed valve, energize to open (D3).** Fail-safe on power loss, at the cost of 8 W of coil power all the time.
- 3. Independent hardware trip (D8).** A comparator with its own latch opens a relay in series with the microcontroller's valve switch, so a firmware bug cannot hold the valve open.
- 4. Continuous ventilation with a boost (D4).** Avoids switching a motor on in a flammable mixture and proves fan rotation before gas can flow.
- 5. Set points 10 % LFL (warn) and 25 % LFL (trip) (D2).** More conservative than the 25 % and 50 % levels often used in industry.
- 6. 24 V DC throughout, with a certified plug-in supply (D9).**

7. **Key-switch reset** (D6). Stops students from clearing a trip without a supervisor.
8. **One detector head per room** (D6). A second head is an option where the ceiling has beams or pockets, or where leak points are far apart (see Key numbers).
9. **Timed escalation** (HGD-DDR-002). A warning held for 5 min closes the valve and latches. Options were (a) this; (b) a lower trip at 20 % LFL; (c) warning only for small leaks. Decided: (a).
10. **Head placement rule** (HGD-DDR-002). A head directly above each likely leak point, with the horizontal offset limit set from the plume width at TRL 4.

■ Relation to other lab projects

- **H2Bench** is the first host. Its README states that it is "Protected by H2Guard", and its gas system (electrolyzer, low-pressure store, regulator and tubing) is where the valve and any flow restrictor go. The inventory rule (R9, D5) applies to H2Bench's storage; its 2 L tank at 300 kPa gauge holds 7.9 L, 2.6 % of the limit. H2Bench's review asks for interlock outputs to cut its power supply and close its tank solenoid; H2Guard has one valve output at present (see the review note).
- H2Guard does not use a SwapCell pack or any lithium cell, and it does not depend on FieldNode, CellGuard, MotionCore, ThermaCart, TwinKit or CalRig.

■ Safety

Safety: Hydrogen is flammable in air from about 4 % to 74 % by volume and ignites with very little energy. H2Guard is a research and teaching prototype, not a certified gas detection system, and it must not be the only safeguard in any room where codes or insurers require certified detection. Use certified equipment for any real installation.

Safety: The parts are not certified for hazardous areas. The catalytic sensor element runs hot, and the fan motor, relays and supply are possible ignition sources. The design limits this by keeping the controller and supply low on the wall, running the fan continuously and fitting flame arrestors to the sensor ports, but it does not remove the risk.

Safety: The design leak assumes a flow restrictor. A failed regulator on an unrestricted high-pressure cylinder can release gas far faster than the fan can remove it. Keep inventories within the R9 limit (300 L at atmospheric pressure for the 30 m³ room) or fit a restrictor, and have gas fittings made and leak-tested by a competent person.

Safety: Catalytic sensors need oxygen and can be poisoned by silicones, sulfur compounds and some solvents, which makes them read low without warning. Bump test before each teaching session and after any exposure to these vapors, and replace the sensor when it fails a bump test.

Safety: A blocked duct or a stuck shutter is not detected while the fan still turns, and a leaking valve seat is not detected at all (HGD-CAL-001, Table 4). Check the airflow and the valve at every session until airflow proving and a valve proof test are designed.

Safety: Mains power enters only through a certified plug-in supply. Do not open it or wire mains inside the controller.

■ Open questions

- Confirm t_{90} , power and poisoning behavior of the chosen catalytic sensor from its datasheet, and whether its hydrogen output gives 1 % LFL resolution (R1, R4).
- Measure the delivery factor of the open bump test cup and set the pass level.
- How airflow is proven: tachometer only, or a differential pressure switch (HGD-DDR-001, O4). Proposed, awaiting Amish.
- Low-cost solenoid valve or a certified gas valve (O5). Proposed, awaiting Amish.
- Where set points and the inventory rule come from for each country the first users are in (O3); oxygen depletion monitoring (O2). Proposed, awaiting Amish.
- Whether a full-span bump test should trip the system or use a lower span gas. Proposed, awaiting Amish (no recommendation until the cup delivery factor is known).
- The horizontal offset limit for the head placement rule, from a measured plume width (TRL 4, on hold).